TIMOTHÉE TEYSSIER

Data Scientist | Machine Learning & Computer Vision

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PROFESSIONAL EXPERIENCE

AI & Computer Vision Engineer — SMP Technik

May 2025 – Present

- Develop industrial state-of-the art AI applications combining **computer vision, machine learning** and software engineering.
- Manage **end-to-end ML workflows**, including dataset annotation, preparation, training and optimization.
- Build multi-stage computer vision pipelines for metal-cutting tool detection, **segmentation and classification**.
- Develop AI models to extract technical characteristics from engineering drawings for machining and sharpening workflows.
- Compare model versions and configurations through iterative experimentation to guide performance improvements.
- **Evaluate** models using precision, recall, F1-score and confusion-matrix analysis, without disclosing confidential results.
- Use appropriate **train-validation-test splits and stratification** to ensure reliable model evaluation.
- Integrate AI inference into desktop and mobile applications, connecting predictions with databases, analysis results and MQTT-based data exchange.

Software Developer — Thelesys

October 2024 – May 2025

- Developed a C++ / Qt application enabling communication between electronic components through UDS services and multithreading, working with real-time data from electronic systems and sensors.
- Worked in a 5-developer **Agile** team.

C++/Qt Software Engineering Intern — Kitware

February 2024 – July 2024

- Contributed to Paralabel, an open-source annotation platform for **large 3D machine-learning datasets**, including unit-test development and validation.
- Applied software development best practices through Git branching, pull requests, code reviews, and version management with tags.
- Project: Paralabel ↗

Medical Imaging Research Intern — Kitware / UNC Chapel Hill

August 2022 – August 2023

- Worked on **on healthcare datasets, including neuronal fiber** classification using deep learning **(PyTorch, Python)**, including model training, experimentation and evaluation, with Dr. Juan Carlos Prieto and Dr. Martin Styner.
- Contributed to DTIPlayground, a medical imaging tool, and developed a Python `Image_Filter` module for filtering **Diffusion Tensor Imaging (DTI)** images.
- GitHub: DTIPlayground ↗

PORTFOLIO · SCAN OR CLICK ↗

KEY SKILLS

AI & MACHINE LEARNING

Python, PyTorch, PyTorch Lightning, Machine Learning, Deep Learning, Supervised Learning, Unsupervised Learning, Contrastive Learning, Dataset Preparation, Model Evaluation, Optimization

COMPUTER VISION

OpenCV, image processing, object detection, segmentation, classification

DATABASES

SQL: PostgreSQL, MySQL, SQLite, NoSQL: MongoDB

DATA ANALYSIS & VISUALIZATION

Statistical Analysis, Plotly, Power BI

SOFTWARE ENGINEERING

C++, Qt, CMake, Git, Docker

TESTING & CI/CD

CTest, pytest, GitLab CI/CD

TOOLS & ENVIRONMENTS

DIPY, VTK, ITK, ONNX, TensorRT, Linux, Windows

EDUCATION

Engineering Degree — CPE Lyon
2020 – 2024
Imaging, Electronics, Robotics and Computer Science with a specialty: (IMI) Image, Modélisation, Informatique

LANGUAGES

French Native

English C1

SOFT SKILLS

Communication: Technical & non-technical audiences

Teamwork: Cross-functional collaboration

Initiative: Project ownership, international presentations

Rigour: Attention to detail, reliable execution

Confidentiality: Sensitive data handling

Problem-solving: Analytical thinking, continuous improvement